Microsoft Elevate

CAPSTONE PROJECT

Smartphone Usage Behavior Analysis Dashboard using Power BI

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Problem Statement:

Smartphones have become an essential part of modern life and significantly influence communication, productivity, and entertainment. However, excessive smartphone usage and increased screen time have raised concerns regarding digital habits and lifestyle balance.

Understanding smartphone usage behavior across different demographic groups such as Age Group, Gender, and Location is difficult without proper data analysis. Important behavioral factors such as Daily Screen Time, Number of Apps Used, and Application Categories including Social Media, Gaming, and Productivity Apps influence overall smartphone usage patterns.

Therefore, there is a need for an interactive data analytics system that can analyze smartphone usage behavior and present meaningful insights using data visualization.

Proposed Solution:

- The proposed solution is to develop an interactive Power BI dashboard that analyzes smartphone usage behavior using a dataset containing user demographic information and application usage data.
- The dataset includes attributes such as Age, Gender, Location, Daily Screen Time Hours, Number of Apps Used, and Application Usage Categories such as Social Media, Gaming, and Productivity Apps.
- The dataset is cleaned and transformed using Power Query, and calculated measures are created using DAX (Data Analysis Expressions).
- Various visualizations such as Bar Charts, Treemaps, Area Charts, and Decomposition Trees are used to analyze usage patterns and identify relationships between different behavioral factors.
- The dashboard also includes interactive slicers and filters that allow users to dynamically explore smartphone usage trends.
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System Approach:

The system is developed using Microsoft Power BI, which provides powerful tools for data transformation, visualization, and business intelligence analysis.
The project requires Power BI Desktop, a computer with sufficient processing capability, and a smartphone usage behavioral dataset containing demographic and mobile usage information.
Power Query is used for data cleaning and preprocessing, while DAX (Data Analysis Expressions) is used to create calculated measures such as average screen time and total app usage.
Using these tools, the dataset is transformed into meaningful visual insights that help analyze smartphone usage behavior effectively.

Algorithm & Deployment:

- In the Algorithm section, describe the machine learning algorithm chosen for predicting bike counts. Here's an example structure for this section:
- **Algorithm Selection:**
- The project uses Power BI data analytics and visualization techniques to analyze smartphone usage behavior and identify patterns across users.
- **Data Input:**
- The dataset includes Age, Gender, Location, Screen Time Hours, Number of Apps Used, and App Usage Categories such as Social Media, Gaming, and Productivity.
- **Data Processing:**
- Data is cleaned using Power Query, and calculated measures like Average Screen Time and Total App Usage are created using DAX.
- **Deployment Process:**
- An interactive Power BI dashboard is developed with charts and filters to explore smartphone usage insights.
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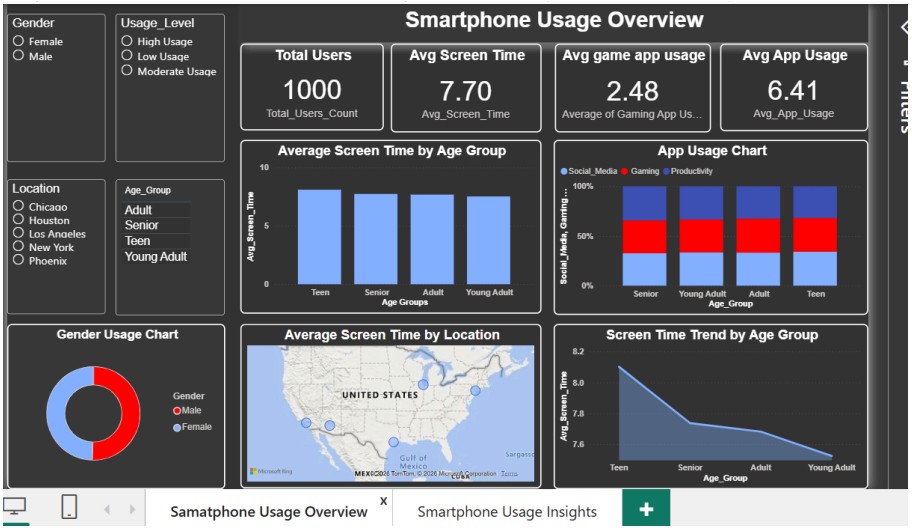
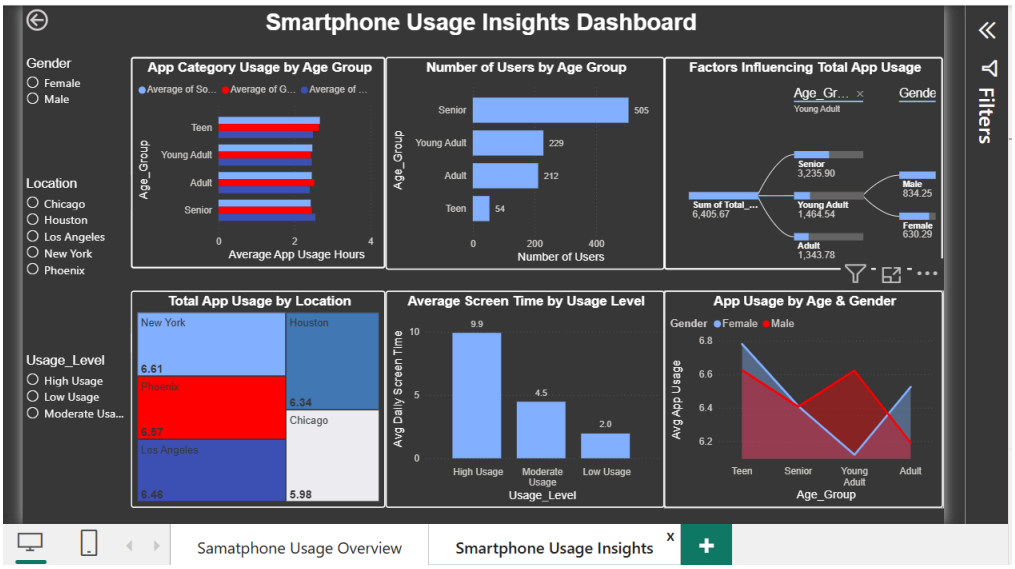
Result:

The developed Power BI dashboard successfully visualizes smartphone usage behavior using multiple interactive charts and analytical visualizations.
The dashboard provides insights into Average Screen Time by Age Group, Application Usage Distribution across Social Media, Gaming, and Productivity Apps, and smartphone usage differences based on Gender and Location.
Advanced visualizations such as Treemap, Decomposition Tree, Area Chart, and Matrix Heatmap help identify important factors influencing smartphone usage behavior.
Users can interact with the dashboard using slicers and filters, allowing them to dynamically explore smartphone usage trends and behavioral patterns

DASHBOARD OUTPUT





Github : [Github link]

Conclusion:

The smartphone usage behavior dashboard provides an effective platform for analyzing and visualizing digital habits across different demographic groups.
Using Power BI data visualization techniques, the project highlights important insights such as screen time trends, application usage patterns, and the influence of factors such as age group, gender, and location.
The interactive dashboard enables users to explore smartphone usage data easily and supports data-driven understanding of modern digital behavior.

Future scope:

The system can be further improved by integrating additional datasets and advanced analytical techniques.
Future improvements may include incorporating real-time smartphone usage data, applying machine learning models to predict smartphone addiction levels, and expanding the analysis to cover multiple geographic regions.
The dashboard can also be integrated into mobile or web-based applications, enabling users to monitor their digital habits and improve digital wellbeing.

References:

References

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4. Data Visualization Concepts and Techniques
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Thank You